

UNIVERSITY OF OSLO

Master's thesis

Topological Quantum Computation with the Kitaev Chain

The Kitaev Chain, Quantum Computation with Non-Abelian Anyons and Numerically Simulating the Kitaev Chain T-Junction

Axl Hendrik Kleven
axlhk@uio.no

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Department of Physics
Faculty of Mathematics and Natural Sciences

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Abstract

This thesis begins by reviewing Kitaevs original article on the Kitaev chain [1], a one-dimensional superconducting chain of spinless fermions. We outline solutions of this model where Majorana zero modes(MZMs) arise at the edges, providing a basis for topologically protected qubits. Further we explore how the Kitaev chain can be used as a qubit, specifically studying the T-chain geometry, showing that this system can implement the Pauli-X and Pauli-Y (and thus Pauli-Z) quantum gates by braiding of the edge MZMs. We also show that this system does not provide the basis for a complete quantum computer, and discuss the missing operations.

We explore methods of numerical simulations of the model, implementing both an adiabatic method allowing for truncation of higher energy states in hopes of speeding up numerical simulations while maintaining accuracy, and direct diagonalization of the full spectrum (for comparison). Initial testing proves to replicate the results of a previous student for the case of cutting a chain, an operation in which a single chain starts with all sites in the topologically non-trivial phase, and then gradually turning the hopping and pairing terms of the central bond to zero, before gradually turning them on again. Further we simulate the Pauli-X and -Y gates, in which two different pairs of majoranas are exchanged, showing that the ground state degeneracy is preserved throughout the operation. For the numerical results we specifically discuss the evolution of the ground state energies for varying total times of the operation T, varying levels of truncation, and the results are heavily restricted in the total number of sites on the chain, at most reaching $N = 13$ sites for a simulation. [Her vil jeg legge inn resultater].

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Chapter 1

Introduction

In 1937 during the founding period of quantum field theory, Ettore Majorana hypothesised the existence of fermions which are their own antiparticle [2]. These became known as Majorana fermions, and were for a long time considered to only be a property of very specific elementary particles, although the existence of such elementary particles is yet to be confirmed. The interest in Majorana fermions in condensed matter physics as quasi particles is a recent phenomenon, bolstered by the surge of interest in topological phases of matter. Originally this interest started in 1980 with the experimental confirmation of the quantum Hall effect [3], which was later theorised to occur due to an arising topological invariant - a Chern number [4]. From this, interest became more widespread with physicists exploring materials characterised by topological phases, like topological insulators [5]. In 2001 Kitaev proposed the Kitaev chain [1]. A one-dimensional superconducting wire with a topological phase which can host Majorana zero modes at the ends of the chain in the form of quasiparticles. Recent experiments have yielded signatures consistent with Majorana zero modes, although there is still strong debate on this topic as to whether or not they were measured with certainty [6].

The Kitaev chain has garnered a lot of attention due to its promise of a topological qubit. As the MZMs are located at the edges of the chain, the overlap of their wavefunctions decreases exponentially with the length of the chain, and their stored information undergoes a topological protection in which decoherence is significantly reduced [7]. In other words a solid theoretical understanding and a following experimental realization of the Kitaev chain could mean opening the door to a decoherence free quantum computer. Such a quantum computer could provide great acceleration in many fields where quantum algorithms such as the famed Grovers[8] or Shor's[9] are applicable, but can also offer exponential speedups in fields in which simulations of many-body quantum Hamiltonians are necessary [10]. A specific example being high-temperature superconductivity [7], with many expected potential applications in broader fields such as medicine, condensed matter, materials science, chemistry, and more.

There are however several issues the field must address for this to become reality. Among others we need a clear scalable method of implementation, measuring and braiding the Majoranas. We need to be able to understand the system at larger system sizes (where we expect the topological protection to be strong) which is currently unfeasible numerically due to the numerical cost increasing exponentially with the Hilbert space. We need a system which is tolerant enough against decoherence to perform about 10^6 operations without errors in order to have sufficient error correction [11]. These are only examples of the theoretical problems that require attention. The engineering

challenges of maintaining a superconductor, preventing quasiparticle poisoning, varying the parameters of the chain to move the MZMs, and more, represent their own challenges that will require marvels of engineering.

This thesis aims to create a solid understanding of the current state of the field, and open doors to further research. First we review the Kitaev chain; study its mathematical and conceptual origins, and then further explore how this system can be used to create a simple topological qubit. Further we will explore the systems theoretical applications, limitations, and properties with regards to a specific architecture, the T-junction. Designed to allow a scalable system of braidable Majoranas. Specifically we will study which operations are available to us in the form of braiding with non-abelian statistics. In continuation of this we will explore a way of numerically simulating the system with hopes of maintaining accuracy and reducing the numerical cost by considering an adiabatic approach and truncating higher energy states. For the results we will also discuss whether or not the braiding operations earlier described are in fact available to us by checking the degeneracy of the ground states throughout the simulation.

Chapter 2

The Kitaev chain

The Kitaev chain is a toy model of a one dimensional spinless p-wave superconductor, and is an example of material that can realize a topological superconducting phase. It consists of a chain of spinless fermions with nearest-neighbour hopping and superconducting pairing, under the influence of a chemical potential. The Hamiltonian takes the form

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{2.1}$$

where a_j and $a_j^\dagger$ are annihilation and creation operators for the fermionic sites $j = 1, 2, \cdots, L$ in the chain, where L (also denoted N for the T-junction later) is the total amount of sites. For the ω, Δ terms we count the bonds, ranging $j = 1, 2, \cdots, L-1$. The parameter ω is the hopping amplitude, μ is the chemical potential, and $\Delta = |\Delta| e^{i\theta}$ is the induced superconducting gap. All of which can be made time dependent. The Hamiltonian serves to give rise to Majorana zero modes (MZMs) at its boundaries. Simply put, these MZMs are theorized to be useful as qubits, and a network of chains can be used for quantum computation.

MZMs are interesting candidates for qubits as their topological nature provides robustness against local perturbations and certain sources of decoherence. Their non-Abelian nature allows for braiding, where the MZMs exchange position and information about the path is stored in the wavefunction. This is the core of the process that allows computation to take place, and we will study this closer in a later chapter 3.

In the first two sections 2.1, 2.2 of this chapter (and this heading) we closely follow the original theory laid out by Kitaev [1], re-deriving the main results and filling in with intermediate steps and discussion for clarity, with an addition of supplementary subsections. More specifically we cover the origin of the MZMs and the phases of the Kitaev chain. We also study BdG formalism and apply it to the Kitaev chain as parts of the phase discussion is based on these results.

2.1 Majorana fermions in the Kitaev chain

In the context of the Kitaev chain, Majorana fermions are quasiparticles which are 'half' of an ordinary fermion and also act as their own antiparticle. The Majorana operators can be defined as

$$c_{2j-1} = a_j + a_j^\dagger, \quad c_{2j} = \frac{a_j - a_j^\dagger}{i}, \quad (j = 1, ..., N) \tag{2.2}$$

which are hermitian and anticommute, meaning they satisfy the relations

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = c_l c_m + c_m c_l = 2\delta_{lm}, \quad (l, m = 1, ..., 2N). \tag{2.3}$$

Using the definitions of the majorana operators 2.2 we can find the number operator for a site j

$$a_j^\dagger a_j = \frac{1}{2}\left(1 + i c_{2j-1} c_{2j}\right), \tag{2.4}$$

where the term $ic_{2j-1}c_{2j}$ corresponds to a phase error. If c_{2j-1} and c_{2j} are not on the same site, but separated in space, then $c_{2j-1}c_{2j}$ would represent an interaction between two separated Majorana sites, requiring the environment to couple the Majorana fermions simultaneously to create the phase error. Physical noise and imperfections are typically local, so placing the Majorana sites far enough away from each other would decrease the risk of a phase error. Further, a single Majorana operator cannot appear by itself in a reasonable Hamiltonian as it would not preserve fermionic parity; flipping the system from an even to odd (or vice versa) number of fermions. As such, the Majorana operators will always come in pairs $c_i c_l$, and thus no parity preserving term can involve just one isolated Majorana. Together this tells us that errors would either have to break parity, or couple both separated Majoranas at once, which is highly unlikely for separated Majorana sites and thus the Majorana edge states are protected from local errors.

2.1.1 Rewriting the Hamiltonian in terms of Majorana operators

We can express the Hamiltonian in terms of Majorana operators. As we will see, this transformation will provide insight about the structure of the phases and the emergence of the edge modes. Before doing so we simplify the calculation by hiding the phase θ in the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{2.5}$$

$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{2.6}$$

We can now perform the transformation of the Hamiltonian by inverting these relations and inserting into the original Hamiltonian 2.1. Together with the hermiticity and anticommutation relations 2.3 we get the Hamiltonian

$$H = \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + (\omega + |\Delta|)\, c_{2j} c_{2j+1} + (-\omega + |\Delta|)\, c_{2j-1} c_{2j+2}\right]. \tag{2.7}$$

This calculation is shown in appendix A.1. Staring at this Hamiltonian, we can first see that it is on the general quadratic Majorana form

$$H = \frac{i}{4} \sum_{l,m} A_{lm} c_l c_m, \quad A^T = -A \tag{2.8}$$

with A being a real and antisymmetric matrix, encoding the intrinsic particle-hole symmetry of the superconducting system. The single particle spectrum is fully determined by the eigenvalues of A, reducing the problem from a 2^L-dimensional many-body operator problem in Fock space to a single-particle linear algebra problem on a $2L \times 2L$ matrix. This is the basis of BdG, as we will get back to later. Further, we see that all direct mentions of the complex phase θ are gone from H. This reflects that θ

can be removed by an appropriate gauge transformation, and is therefore irrelevant for the bulk spectrum and the topological classification of the phases, which we will also touch upon later.

The Majorana representation also gives us a geometric picture of the chain. The Hamiltonian now consists of three terms, providing three different types of couplings. An on-site term $-\mu c_{2j-1}c_{2j}$, and two nearest-neighbour terms $(\omega + |\Delta|)c_{2j}c_{2j+1}$ and $(-\omega + |\Delta|)c_{2j-1}c_{2j+2}$. The on-site term couples two Majorana modes per site j, so for a chain of length L we get $2L$ Majorana operators. We can garner further insight by studying specific cases for the parameters μ, ω and $|\Delta|$ of the Hamiltonian.

Topologically trivial case $\omega = |\Delta| = 0, \mu > 0$

In the case where μ dominates the Hamiltonian reduces to

$$H = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) = \frac{i}{2} \mu \sum_j c_{2j-1} c_{2j}. \tag{2.9}$$

In this case the system is topologically trivial as the Majoranas on each site are glued together to form a complex fermion. There are no unpaired Majorana fermions at the edges of the chain, making this case uninteresting for further study.

Topologically non-trivial case $\omega = |\Delta| > 0, \mu = 0$

In the case where the hopping and superconducting parameters dominate the Hamiltonian reduces to

$$H = i \sum_j \omega c_{2j} c_{2j+1}, \tag{2.10}$$

where the nearest neighbours between different sites dominate. This is an interesting case as c_2 couples to c_3, c_4 couples to c_5, and so on, meaning the majorana operators from different sites pair together. But c_1 and c_{2L}, the Majorana edge states, do not appear in the Hamiltonian. They have however not disappeared physically, but rather remain as zero-energy Majorana modes localized at the edges of the wire. Together they form a nonlocal fermionic zero mode. This is the most important insight from the transformation of the Hamiltonian as these are the states we were looking to create.

The two cases above can be thought of by a common intuitive picture of a chain of people with two hands. The people represent the sites, and their hands represent the two majorana fermions associated with the site. In the trivial case everyone is holding their own hands, representing how the majorana fermions have paired together at the same site. Whereas in the topologically non-trivial case the people have let go of their own hands, and are now holding hands with their neighbours, representing how the majorana fermions are now paired with the majorana of the neighboring site. Now there are suddenly two open hands: one at each end, representing the two topologically separated majoranas.

We can now define new annihilation and creation operators spanning the sites j and $j+1$

$$d_j = \frac{1}{2} \left(c_{2j} + i c_{2j+1} \right) \tag{2.11}$$

$$d_j^\dagger = \frac{1}{2} \left(c_{2j} - i c_{2j+1} \right). \tag{2.12}$$

Such that we can rewrite the Hamiltonian

$$H = 2\omega \sum_{j=1} \left(f_j^\dagger f_j - \frac{1}{2} \right). \tag{2.13}$$

As the edge states are not mentioned in the Hamiltonian, this expression fully determines the energy spectrum of the bulk. We see that the bulk consists of fermionic modes with energy $\propto 2\omega$. Further, the ground states satisfy the condition $d_j|\psi\rangle = 0$ for all j. As such, all these modes will remain empty, and any ground state $|\psi\rangle$ is fully determined by the edge majoranas c_1 and c_{2L}. Therefore we get two orthogonal states where the above ground state condition for the bulk is true; $|\psi_0\rangle$ and $|\psi_1\rangle$. These ground states will be eigenstates of the hermitian operator $-ic_1c_{2L}$ as the edge Majoranas represent unpaired degrees of freedom. This operator also squares to 1, and as such has eigenvalues ± 1. We get

$$-ic_1c_{2L}|\psi_0\rangle = |\psi_0\rangle, \quad -ic_1c_{2L}|\psi_1\rangle = -|\psi_1\rangle, \tag{2.14}$$

where the labeling of the states were chosen arbitrarily. We see that the total fermion parity operator

$$P = \prod_j (-ic_{2j-1}c_{2j}) \tag{2.15}$$

has the same contribution from the bulk for both ground states. So the difference in parity is solely a result of the edge contribution $-ic_1c_{2L}$. The even parity of $|\psi_0\rangle$ represents the state being a superposition of states with an even number of electrons. And $|\psi_1\rangle$ has an odd parity, representing an odd number of electrons. Together they reflect the boundary degrees of freedom from the unpaired Majorana modes at the edges.

Arbitrary values of ω, μ, Δ

For arbitrary values of ω, μ, Δ the Hamiltonian has the general quadratic form (2.8) in terms of the Majorana operators. This means the Hamiltonian can be diagonalized into independent fermionic modes by a change of basis, implemented by a matrix W. We require that W is a $2N \times 2N$ real orthogonal ($W^TW = WW^T = I$) matrix, and that it has two further properties; the first being that it defines new operators $\{b'_m, b''_m\}$ by the linear transformation $b = Wc$, or

$$\begin{pmatrix} b'_1 \\ b''_1 \\ \vdots \\ b'_N \\ b''_N \end{pmatrix} = W \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_{2N-1} \\ c_{2N} \end{pmatrix}. \tag{2.16}$$

This property is ensured by W being real and orthogonal, such that after the transformation the Majorana anticommutation relations are preserved. This can be

shown using $b_\gamma = \sum_\alpha W_{\gamma\alpha} c_\alpha$, where $W_{\gamma\alpha}$ are just real numbers, meaning they commute

$$\begin{aligned}
\{b_\gamma, b_\delta\} &= \sum_{\alpha,\beta} W_{\gamma\alpha} W_{\delta\beta} \{c_\alpha, c_\beta\} && (2.17)\\
&= \sum_{\alpha,\beta} W_{\gamma\alpha} W_{\delta\beta} 2\delta_{\alpha\beta} && (2.18)\\
&= 2\sum_{\alpha} W_{\gamma\alpha} W_{\delta\alpha} && (2.19)\\
&= 2\sum_{\alpha} W_{\gamma\alpha} W^T_{\alpha\delta} && (2.20)\\
&= 2\delta_{\gamma\delta} && (2.21)
\end{aligned}$$

where the last step is a result of W being orthogonal. The second property stems from the antisymmetry of A, such that in this basis A acquires a canonical block-diagonal form

$$WAW^T = \begin{pmatrix} 0 & \epsilon_1 & & & \\ -\epsilon_1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & \epsilon_N \\ & & & -\epsilon_N & 0 \end{pmatrix}. \tag{2.22}$$

This condition stems from a mathematical theorem on the canonical form of real antisymmetric matrices, which states that if we have a real antisymmetric matrix A of size $2N \times 2N$ there exists a real orthogonal matrix W such that A can be transformed into a block-diagonal form

$$WAW^T = \mathrm{diag}(C_1, C_2, \cdots, C_N), \quad C_j = \begin{pmatrix} 0 & -\lambda_j \\ \lambda_j & 0 \end{pmatrix} \tag{2.23}$$

with $\lambda_j > 0$ being real numbers, stemming from the pairs of complex eigenvalues $\pm i\lambda_j$ for A. This theorem is a standard result in matrix analysis, and a proof can be found in [12] [1]. Once we have a matrix W that satisfies the above conditions we can reduce the Hamiltonian further to a canonical form by inverting the transformation $b_\gamma = \sum_\alpha W_{\gamma\alpha} c_\alpha$

[1] There are apparently many books on this topic, none of which i had access to. This is the best source i could find without proving it directly.

to $c_\alpha = \sum_\gamma W^T_{\alpha\gamma} b_\gamma = \sum_\gamma W_{\gamma\alpha} b_\gamma$

$$H_{\text{canonical}} = \frac{i}{4} \sum_{\alpha,\beta} A_{\alpha\beta} c_\alpha c_\beta \tag{2.24}$$

$$= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} A_{\alpha\beta} W_{\gamma\alpha} b_\gamma W_{\delta\beta} b_\delta \tag{2.25}$$

$$= \frac{i}{4} \sum_{\alpha,\beta,\gamma,\delta} W_{\gamma\alpha} A_{\alpha\beta} W_{\delta\beta} b_\gamma b_\delta \tag{2.26}$$

$$= \frac{i}{4} \sum_{\gamma,\delta} \left(W A W^T \right)_{\gamma\delta} b_\gamma b_\delta \tag{2.27}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \begin{pmatrix} b'_m & b''_m \end{pmatrix} \begin{pmatrix} 0 & \epsilon_m \\ -\epsilon_m & 0 \end{pmatrix} \begin{pmatrix} b'_m \\ b''_m \end{pmatrix} \tag{2.28}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m \left(b'_m b''_m - b''_m b'_m \right) \tag{2.29}$$

$$= \frac{i}{4} \sum_{m=1}^{N} \epsilon_m 2 b'_m b''_m \tag{2.30}$$

$$= \frac{i}{2} \sum_{m=1}^{N} \epsilon_m b'_m b''_m = \sum_{m=1}^{N} \epsilon_m \left(\tilde{a}^\dagger_m \tilde{a}_m - \frac{1}{2} \right) \quad (\epsilon_m \geq 0). \tag{2.31}$$

Where we used that $\{b'_m, b''_m\} = 0$ as they are separate components of b. In this form b'_m and b''_m are real linear combinations of c_{2j-1} and c_{2j}, retaining the commutation relations. N is the number of fermionic sites in the system, here $N = L$. The complex fermion quasiparticles are defined as $\tilde{a}_m = \frac{1}{2}\left(b'_m + i b''_m\right)$ and $\tilde{a}^\dagger_m = \frac{1}{2}\left(b'_m - i b''_m\right)$.

The key take-away here is that the Kitaev chain Hamiltonian, due to being quadratic, can be diagonalized into independent quasi particles $\tilde{a}_m$ with energies ϵ_m. This is how the problem can be reduced from a 2^L-dimensional many-body problem to single-particle energies and modes on a $2L \times 2L$ matrix using BdG formalism as we will later cover 2.3.

2.1.2 Properties of Majorana fermion quasiparticles

Let us take a minute to think about what a Majorana fermion quasiparticle really is. As the previous subsection also started by stating, a majorana fermion is a particle that is its own antiparticle. In other words, it obeys

$$c = c^\dagger. \tag{2.32}$$

At first glance[2] it might appear to be a fairly tame statement. This is however a rather weird statement which gives the Majorana fermions interesting properties. This equation essentially enforces that applying a majorana fermion annihilation operator is equivalent to applying a creation operator: removing a majorana fermion is mathematically identical to creating one.

This completely changes how we need to think about the Majorana fermion compared to a Dirac fermion. For Dirac fermions, when they are brought close, the Pauli principle sets in and keeps them apart. All combinations of Majorana raising and lowering operators however obey the anticommutation relation as seen in 2.3, which does not

[2] In fact, 6 months of first glances for some

enforce the Pauli exclusion principle for same states, it instead enforces $(c_l^\dagger)^2 = 1$. The anticommutation relation also however enforces $c_l^\dagger c_m^\dagger = -c_m^\dagger c_l^\dagger$ for $m \neq l$, implying the same exchange statistics for MFs as for Dirac Fermions. Furthermore, the number operator $n = c_i^\dagger c_i = c_i^2 = 1$ does not make sense when looking exclusively at the MFs. Remember however that the Majorana fermions are quasi-particles which emerge from Fermionic sites a_i, and we have re-defined the sites using Majorana operators in (2.12), which we called d_i. From this we can construct a meaningful number operator

$$n_i = d_j^\dagger d_j = \frac{1 + \gamma_1 \gamma_2}{2}, \tag{2.33}$$

which has eigenvalues $n_i = 0, 1$. This number operator is experimentally measurable by bringing two MFs closer together and measuring the energy of the state. The energy will be caused by the overlap of the MFs and is described by the effective Hamiltonian

$$H_{\text{eff}} = \frac{i}{2} t c_{2i-1} c_{2i} = t \left(n_i - \frac{1}{2} \right), \tag{2.34}$$

which we will later discuss in more detail in 2.2.3, where we also detail the strength of the overlap t. When two MFs are brought even closer and collide, (2.32) seems to imply that the MFs annihilate. We can however see that they instead pair up to form an ordinary Dirac fermionic site with an energy defined by this Hamiltonian.

Another property of MFs, perhaps obvious from the original derivation, is that they always come in pairs. A system has no trouble having an odd or even number of Dirac fermions, but there will always be an even number of MFs. The MFs can be separated in space, and so if a single MF appears somewhere in the system, another MF must appear somewhere else in the system. [13] [14]

These are properties which typically arise from the particle-hole symmetry of superconducting systems, and is in fact an important reason as to why superconductors serve as a hunting ground for Majorana fermions. We will not be discussing the relevant properties of superconductors further [3]. A good introduction to MFs in superconductors can be found in [7].

2.2 Phases of the Kitaev chain

The Kitaev chain is really only interesting to us due to its promise of MZMs at its edges. We have previously seen that these quasiparticles only appear for specific parameters. In other words the Kitaev chain can host phases where the MZMs are, or are not present. Previously we have referred to these phases as the topologically non-trivial and the topologically trivial phase, respectively. The goal of this section is to further our understanding of which relation of parameters causes the chain to be in which phase, with a specific focus on the distinction between the topological and the trivial phase.

2.2.1 Phases of the bulk spectrum

The bulk spectrum for energy vs. momentum q for a chain of infinite length is given by

$$\epsilon(q) = \pm \sqrt{(2\omega \cos(q) + \mu)^2 + 4|\Delta|^2 \sin^2(q)}, \quad -\pi \leq q \leq \pi. \tag{2.35}$$

Finding this relation is done by Fourier transforming the Hamiltonian to momentum space as shown in appendix A.2 and then finding its eigenvalues, as we will show in

[3] Although an attempt was made

2.3. This was done using periodic boundary conditions as we expect the same result when $L \to \infty$ for a chain. For now you can take this for granted as its derivation gives little insight into the physics of the system. This energy spectrum will tell us where the system is gapped ($\epsilon(q) \neq 0$), and where gap closings ($\epsilon(q) = 0$) occur. The gap closings mark boundaries between topological phases [4]. We now find the gap closings

$$\epsilon(q) = 0 \tag{2.36}$$

$$\pm\sqrt{(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q)} = 0 \tag{2.37}$$

$$(2\omega\cos(q) + \mu)^2 + 4|\Delta|^2\sin^2(q) = 0 \tag{2.38}$$

Both terms in this equation are real values squared, and must therefore be nonnegative. They must also both individually vanish to fulfill the equation, and we thus get two equations

$$2\omega\cos(q) + \mu = 0, \quad \sin(q) = 0. \tag{2.39}$$

From the second equation we get $\sin(q) = 0$ for $q = 0, \pi$. Plugging this into the first equation we get

$$\begin{aligned} 2\omega\cos(0) + \mu = 0 \qquad & \qquad 2\omega\cos(\pi) + \mu = 0 \\ 2\omega = -\mu \qquad & \qquad 2\omega = \mu \end{aligned} \tag{2.40}$$

Meaning the bulk gap closes when $2|\omega| = |\mu|$, defining a phase boundary between the topologically trivial and non-trivial superconducting phases.

From our previous findings we know that the system will be in the topologically non-trivial phase when $\mu < \omega$ (from the case $\omega = |\Delta| > 0, \mu = 0$), which is in the domain $|\mu| < 2|\omega|$. We here expect that if the parameters change without crossing the bulk gap boundary $2|\omega| = |\mu|$ the system should remain in the same phase. Further we need $\Delta \neq 0$ to remain in the superconducting phase. Meaning the phases of the system can be divided into

$2|\omega| > |\mu|, \Delta \neq 0$: Topologically non-trivial superconductor (MZMs).
$2|\omega| = |\mu|, \Delta \neq 0$: The superconducting bulk gap closes (phase boundary).
$2|\omega| < |\mu|, \Delta \neq 0$: Topologically trivial superconductor (no MZMs).
$2|\omega| > |\mu|, \Delta = 0$: Gapless metal.
$2|\omega| < |\mu|, \Delta = 0$: Gapped insulator.

2.2.2 Boundary modes

In the previous subsection we expected the phase of the system to remain the same as long as the bulk gap boundary is not crossed. We can check that this is true by studying the boundary modes, as they correspond to eigenvectors of A localized near the ends of the chain, or in other words can confirm whether or not the phase is in the topological phase containing Majorana fermions. Such zero modes should be real linear combinations of c_{2j-1}, c_{2j}, having the form

$$b' = \sum_j \beta'_j c_{2j-1}, \quad b'' = \sum_j \beta''_j c_{2j} \tag{2.41}$$

[4] For a more detailed dive into how the closings of the bulk gap signify topological phase transitions see Sec. II.A in [15]

where b' and b'' should be localized on opposite sides of the chain. Here β'_j, β''_j are real coefficients of the Majorana operators. If b' and b'' are zero modes they have no energy and should commute with the Hamiltonian $[H, b'] = [H, b''] = 0$. Further, because the Hamiltonian only couples nearest neighbours the matrix A will be tridiagonal with general values a, b, c, meaning the bulk (excluding the edge modes) can be expressed by

$$a'\beta'_{j+1} + b'\beta'_j + c'\beta'_{j-1} = 0, \tag{2.42}$$
$$a''\beta''_{j+1} + b''\beta''_j + c''\beta''_{j-1} = 0. \tag{2.43}$$

The values a, b, c for their respective β will follow from the Hamiltonian; b being the on-site term μ, while a and c will be the off-diagonal pairing terms ($\omega \pm |\Delta|$). These follow from the commutation relations mentioned above, and will allow us to find solutions for b', b''. We will here show the calculation for b', and show the final result of b'', which has a similar calculation.

$$[H, b'] = \sum_{l=1}^{L} \beta'_j \, [H, c_{2l-1}] \tag{2.44}$$
$$= \sum_{l=1}^{L} \beta'_j \, (i\mu\delta_{jl}c_{2j} + i(\omega + |\Delta|)\delta_{l,j+1}c_{2j} - i(-\omega + |\Delta|)\delta_{lj}c_{2j+2}) \tag{2.45}$$
$$= i\sum_{j=1}^{L} \mu\beta'_j c_{2j} + i\sum_{j=1}^{L-1} \beta'_{j+1}(\omega + |\Delta|)c_{2j} - \sum_{j=2}^{L-1} \beta'_{j-1}(-\omega + |\Delta|)c_{2j} \tag{2.46}$$
$$[H, b''] = -i\sum_{j=1}^{L} \mu\beta''_j c_{2j-1} - i\sum_{j=2}^{L-1} \beta''_{j-1}(\omega + |\Delta|)c_{2j-1} + i\sum_{j=1} \beta_{j+1}(-\omega + |\Delta|)c_{2j-1}. \tag{2.47}$$

Where we used $\{c_m c_n, c_l\} = 2(\delta_{ln}c_m - \delta_{lm}c_n)$, crossed out terms where $\delta_{l+1,j}$ will always be 0, and redefined $j = j + 1$ in the c_{2j+2} term to make all terms c_{2j}. These terms are general. In the bulk, the three sums can be combined into one as we can ignore the boundary cases $j = 1$ and $j = L$. And at the boundaries one of the three terms will be irrelevant (as we would have a missing site), but the statement $[H, b'] = [H, b''] = 0$ still holds true and gives the boundary conditions. Our problem now reduces to finding β_j that are localized near a boundary and satisfy the above equations. Looking at the bulk we see that we can try solutions on the form $\beta'_j = x^j, \beta''_j = x^{-j}$, and that the equations then will reduce to the same characteristic equation

$$(\omega + |\Delta|)x^{j+1} + \mu x^j + (\omega - |\Delta|)x^{j-1} = 0 \tag{2.48}$$
$$(\omega + |\Delta|)x^2 + \mu x + (\omega - |\Delta|) = 0 \tag{2.49}$$
$$\rightarrow x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4|\Delta|^2}}{2(\omega + |\Delta|)}. \tag{2.50}$$
$$\tag{2.51}$$

From this the general solutions follow on the form

$$\beta'_j = \alpha'_+ x_+^j + \alpha'_- x_-^j, \tag{2.52}$$
$$\beta''_j = \alpha''_+ x_+^{-j} + \alpha''_- x_-^{-j}. \tag{2.53}$$

With $\alpha_\pm$ being constants related to $x_\pm$. Therefore we can write an ansatz for the zero modes on either side of the chain

$$b' = \sum_j \left(\alpha'_+ x_+^j + \alpha'_- x_-^j\right) c_{2j-1} \tag{2.54}$$

$$b'' = \sum_j \left(\alpha''_+ x_+^{-j} + \alpha''_- x_-^{-j}\right) c_{2j} \tag{2.55}$$

We have previously confirmed that the topological phase boundary lies at $2|\omega| = |\mu|, \Delta \neq 0$, with expected majoranas at (a) $2|\omega| > |\mu|, \Delta \neq 0$, and no expected majoranas at (b) $2|\omega| < |\mu|, \Delta \neq 0$. We therefore want to check the boundary modes of these two phases to ensure that this is indeed correct, and we remain in the expected topological phase within these domains.

Before doing so we must clarify our expectations. If the zero mode b' denotes a zero mode on the left side of the chain, then b'' will denote a zero mode on the right side (or vice versa). This is due to the exponential form x^j: If $|x_\pm| < 1$, then $x_\pm^j \to 0$ as j increases, causing the zero mode b' to be localized at the first site, $j = 1$. Whereas $x_\pm^{-j} \to \infty$ as j increases, which after normalization causes b'' to be localized at the end of the chain (again vice versa for $|x_\pm| > 1$). As such, we expect that when both $|x_\pm| < 1$ or $|x_\pm| > 1$ we have localized zero modes, with the second case causing them to have opposite orientations.

If however one solution is smaller than 1, and the other is larger, $|x_\pm| < 1, |x_\mp| > 1$, we see that the zero modes b', b'' become a sum of one element approaching 0, and another approaching ∞. In other words, one component is 'localized' at an end, while the other is growing into the bulk. This is not a localized zero mode. If this were to produce a localized zero mode at one end of the chain, we'd need one of the $\alpha_\pm = 0$. This however breaks with boundary conditions. If we check the simplest missing state at the end of a chain $j = 0$ for b' we see that $\beta'_0 = \alpha'_+ + \alpha'_- = 0$, which is impossible to satisfy if one $\alpha_\pm = 0$, and the other $\alpha_\mp \neq 0$. Similar arguments can be made for both ends of the chain for both b', b''. As such, we expect the case $|x_\pm| < 1, |x_\mp| > 1$ to produce no localized zero modes.

Figuring out when $|x_\pm|$ is larger than or smaller than 1 is not directly obvious, but it can be done with a mathematical trick. If we cast $|x_\pm|$ into the complex plane and setting $|x_\pm| = 1 = e^{i\theta}$ to be the unit sphere, we can figure out the conditions on the parameters when $|x_\pm|$ crosses the unit sphere. Plugging this into our characteristic equation 2.49 we find

$$(\omega + |\Delta|)\, e^{i\theta} + \mu + (\omega - |\Delta|)\, e^{-i\theta} = 0 \tag{2.56}$$

$$2\omega \cos(\theta) + 2i|\Delta| \sin(\theta) + \mu = 0 \tag{2.57}$$

Where the complex and the real terms have to equal 0 separately, meaning we require $\sin(\theta) = 0 \to \theta = 0, \pi$, giving the solution

$$2\omega = \pm\mu \tag{2.58}$$

$$2|\omega| = |\mu|. \tag{2.59}$$

Meaning that the value of $|x_\pm|$ can only cross 1 when this condition is met[5]. As such, we can study the domains above and below this condition to determine which topological phase the domain belongs to, and we will remain in this phase if this boundary is not crossed.

[5]It is perhaps unsurprising and definitely reassuring that this is the same result as earlier

(a) $2|\omega| < |\mu|, \Delta \neq 0$

From our condition above we see that we can safely consider the case $\omega = |\Delta|$, which significantly simplifies the problem

$$x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4\omega^2}}{4\omega} \tag{2.60}$$

$$= \frac{-\mu \pm \mu}{4\omega} \tag{2.61}$$

$$\rightarrow x_- = \frac{-\mu}{2\omega}, \quad x_+ = 0 \tag{2.62}$$

$$\rightarrow |x_-| < 1, \quad |x_+| > 1. \tag{2.63}$$

If we were to use different signs for the variables the inequalities would simple be reversed, but never the same for both $|x_\pm|$. As seen from our discussion above this corresponds to the zero modes not existing in this case.

(b) $2\omega > |\mu|, \Delta \neq 0$

If we again consider $\omega = |\Delta|$ we find

$$x_\pm = \frac{-\mu \pm \sqrt{\mu^2 - 4\omega^2 + 4\omega^2}}{4\omega} \tag{2.64}$$

$$= \frac{-\mu \pm \mu}{4\omega} \tag{2.65}$$

$$\rightarrow x_- = \frac{-\mu}{2\omega}, \quad x_- = \frac{0}{2\omega} \tag{2.66}$$

$$\rightarrow |x_-| < 1, \quad |x_+| < 1. \tag{2.67}$$

In this case we have Majorana zero modes b' localized near $j = 0$, and b'' near $j = L$. And if we change the sign convenction to $-\omega > |\mu|$, we simply flip the position of b' and b''. This confirms our initial expectation that the zero modes will remain in the system when the parameters are changed, as long as $|\Delta| \neq 0$ and we do not cross the boundary from the domain $2|\omega| > |\mu|$ to $2|\omega| < |\mu|$.

2.2.3 Overlap of the Majorana wave functions

This subsection might have been better placed in the previous section on Majorana fermions. It is however dependent on the previous discussion on boundary modes, and is therefore placed here for a more pedagogical flow.

Up until this point, the discussion has really been for either a periodic chain, or a chain with $L \rightarrow \infty$. But in a realistic system, we'd have a finite L. Recalling the discussion above, the localized MZMs are actually not actually perfectly localized, but rather the coefficients of the majorana operators decay exponentially fast into the bulk in the topological phase. So for a system with finite L there would be some overlap between the MZMs which would decrease exponentially with the length of the chain. This overlap would cause some coupling between the MZMs, and we would have a non-zero energy based on the overlap. Seeing as j is real and $|x_\pm| < 1$, we can describe the tail of the coefficients in the bulk as

$$|x|^j = e^{j \ln |x|} \tag{2.68}$$

$$= e^{-j|\ln |x||} \tag{2.69}$$

$$= e^{-jl_0}. \tag{2.70}$$

This will be governed by whichever root decays the most slowly, so we define $l_0^{-1} = \min(|\ln|x_-||, |\ln|x_+||)$. The tail of the leftmost MZM would go as $e^{-j|ln|x_\pm||}$ whereas the tail of the rightmost MZM would go as $e^{-(L-j)|ln|x_\mp||}$. Multiplying them together we find the overlap $t \propto e^{-L/l_0}$. Using this, we can describe the interaction by an effective Hamiltonian on the form

$$H_{\text{eff}} = \frac{i}{4}\sum_{l,m} A_{lm} b_l b_m \tag{2.71}$$

$$= \frac{i}{4}\left(tb'b'' - tb''b'\right) \tag{2.72}$$

$$= \frac{i}{4}\left(tb'b'' + tb'b''\right) \tag{2.73}$$

$$= \frac{i}{2} tb'b'' \tag{2.74}$$

$$= t(d^\dagger d - \frac{1}{2}). \tag{2.75}$$

This will cause an energy splitting between the ground states, and they will be split by an energy t. A high t would break the degeneracy of the ground states, causing great problems for our topological protection and quantum computation schemes as a whole. As such, figuring out which values of t are necessary to for the system to be in a degenerate ground state, maintaining topological protection and allowing quantum computation is an important aspect of the field. The most obvious way to maintain a low t is simply by making the chain longer, separating the MZMs. Another way of lowering t is by making the tail $|x|$ decay faster, which can be done by minimizing $|x|$ in 2.50 with respect to the parameters μ, ω, Δ, while remaining in the topological phase of course.

2.3 BdG

BdG formalism can also offer great reductions in the dimensions of the problem, and is as such a common formalism throughout the field.

2.3.1 General BdG form and Nambu basis

"the objective of the transformation is to cast a quadratic Hamiltonian into a diagonal form that resembles the Hamiltonian of a system of non-interacting particles." - [16]

2.3.2 BdG form of the Kitaev chain Hamiltonian

We want to express the Hamiltonian in momentum space as

$$H = \frac{1}{2}\sum_q \Psi_q^\dagger \mathcal{H}_{\text{BdG}} \Psi_q, \tag{2.76}$$

using the Nambu basis (spinors)

$$\Psi_q^\dagger = \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix}, \quad \Psi_q = \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix}. \tag{2.77}$$

Where the momentum-space fermioninic annihilation and creation operators $a_q, a_q^\dagger$ are defined by the fourier transformation

$$a_j = \frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger. \tag{2.78}$$

Where $q = \frac{2\pi n}{N}$ with n such that q runs over the Brillouin zone $[-\pi, \pi]$. We see that $\mathcal{H}_{\text{BdG}}$ has to be a 2×2 matrix. If we look at an arbitrary $\mathcal{H}_{\text{BdG}}$ we can see that after multiplication with the Nambu spinors the Hamiltonian becomes

$$H = \frac{1}{2} \sum_q \begin{pmatrix} a_q^\dagger & a_{-q} \end{pmatrix} \begin{pmatrix} a(q) & b(q) \\ c(q) & d(q) \end{pmatrix} \begin{pmatrix} a_q \\ a_{-q}^\dagger \end{pmatrix} \tag{2.79}$$

$$= \frac{1}{2} \sum_q \left(a(q) a_q^\dagger a_q + b(q) a_q^\dagger a_{-q}^\dagger + c(q) a_{-q} a_q + d(q) a_{-q} a_{-q}^\dagger \right). \tag{2.80}$$

This should match the momentum-space transformed Hamiltonian as found in A.2. We therefore rewrite the Hamiltonian to match the above description. We use the anti-commutation relations of the operators together with $q = -q$ from the periodic boundary conditions, and drop inessential constants

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q - i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{2.81}$$

$$= \frac{1}{2} \sum_q \left[2 \left(-\mu - 2\omega \cos(q) \right) a_q^\dagger a_q - 2i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{2.82}$$

$$= \frac{1}{2} \sum_q \left[\left(-\mu - 2\omega \cos(q) \right) a_q^\dagger a_q + \left(-\mu - 2\omega \cos(q) \right) a_q^\dagger a_q - 2i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{2.83}$$

$$= \frac{1}{2} \sum_q \left[\left(-\mu - 2\omega \cos(q) \right) a_q^\dagger a_q + \left(\mu + 2\omega \cos(-q) \right) a_{-q} a_{-q}^\dagger - 2i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right]. \tag{2.84}$$

We see that we must have

$$a(q) = -\mu - 2\omega \cos(q) \tag{2.85}$$

$$b(q) = -2i\Delta \sin(q) \tag{2.86}$$

$$c(q) = 2i\Delta^* \sin(q) \tag{2.87}$$

$$d(q) = \mu + 2\omega \cos(-q) = \mu + 2\omega \cos(q) \tag{2.88}$$

$$\rightarrow \mathcal{H}_{\text{BdG}} = \begin{pmatrix} -\mu - 2\omega \cos(q) & -2i\Delta \sin(q) \\ 2i\Delta^* \sin(q) & \mu + 2\omega \cos(q) \end{pmatrix}. \tag{2.89}$$

Diagonalizing the BdG Hamiltonian

By diagonalizing the BdG Hamiltonian we can find the energy spectrum. We find its eigenvalues by

$$\det \left(\mathcal{H}_{\text{BdG}} - \epsilon I \right) = 0 \tag{2.90}$$

$$\tag{2.91}$$

which by the abc-formula gives the energy spectrum

$$\epsilon = \pm \sqrt{\left(\mu + 2\omega \cos(q) \right)^2 + 4|\Delta|^2 \sin^2(q)} \tag{2.92}$$

Chapter 3

Quantum computation

The main idea for using Majorana fermions for quantum computation is through a process called braiding. In this process the Majoranas are spatially rotated around eachother, producing a difference in phase. In 1D this is however hard to do: If two Majoranas meet, they simply combine into a normal fermion, allowing the stored information to decohere. Several methods are being discussed to overcome this obstacle, and we will closer study especially one of them; the T-junction. In this junction we will show that one can perform the Pauli X and Z gates (and therefore also Y) but we still need an extra gate, not provided by topologically protected operations, in order to create a general quantum computer.

"A topological quantum computer is constructed using a system in a non-Abelian topological phase. A computation is performed by creating quasiparticles, braiding them, and measuring their final state. " [7]

3.1 Logical qubits from Majorana modes

3.1.1 Degenerate ground states

Recalling the previously discussed Kitaev chain Hamiltonian, and the following definition using the Majorana operators c_m with properties

$$d_j = \frac{1}{2}\left(c_{2j} + ic_{2j+1}\right) \tag{3.1}$$

$$d_j^\dagger = \frac{1}{2}\left(c_{2j} - ic_{2j+1}\right) \tag{3.2}$$

$$c_m^\dagger = c_m, \quad \{c_l, c_m\} = 2\delta_{lm}. \tag{3.3}$$

We recall that in the trivial phase, Majoranas on all sites pair up into ordinary fermions with non-zero energy. However in the topological phase the chain experiences unpaired Majoranas at the edges, which we want well-separated (meaning their wavefunctions overlap negligibly). For a 1D wire of $2M$ such edge Majoranas one can generally construct

M fermion operators f_{lm} [1] with the properties

$$f_{lm} = \frac{1}{2}\left(c_l + ic_m\right) \tag{3.4}$$

$$\{f_{lm}, f_{lm}^\dagger\} = 1 \tag{3.5}$$

$$f_{ij}^2 = \left(f_{ij}^\dagger\right)^2 = 0 \tag{3.6}$$

These correspond to zero-energy modes, and can give rise to 2^M degenerate ground states which can be labeled by occupation numbers $n_j = f_j^\dagger f_j = 0, 1$.

The MZMs need to remain degenerate throughout the operation otherwise they will lose information (? - ref. Joakim viste noe med en energi avhengig fase for majoranaene $|0_L\rangle = e^{\phi_1(E_1)}|\gamma_1\rangle + e^{\phi_2(E_2)}|\gamma_2\rangle$)

3.1.2 Logical basis states

Four Majoranas - a proper qubit

If we have a single Kitaev chain in the topological phase we can produce 2 Majorana zero mdoes.

Conservation of global fermion parity P - Fixed P means that no pair can encode a qubit

In the case of having two wires in the topological phase we can produce 4 Majorana zero modes, which can be used to create logical basis states

$$|0_L\rangle = |0_{12}, 0_{34}\rangle \tag{3.7}$$

$$|1_L\rangle = |1_{12}, 1_{34}\rangle = f_{12}^\dagger f_{34}^\dagger |0_L\rangle \tag{3.8}$$

$0_L = 0, 0$
Pauli operators

3.1.3 Nonlocality and wavefunction picture

Overlap decays as exp(-L / xi)
[17]
[15]

3.2 Braiding and non-abelian statistics (exchange of Majoranas)

nu = 5/2 Qubits and Gates P.52 [7] - Finn koblingen nu=5/2 til Kitaev chain.

3.2.1 Non-abelian anyons

Leinaas

3.2.2 General result

Berry phase
Tegning av braiding?

[1] Note here that f_{lm} is a special case of the the general d_j shown above, only f_{lm} is specifically for the Majorana edge modes

Figure 3.1

Figure 3.2

3.2.3 Clifford group

Exchange operator R [18]

X, Z, Y = iXZ

3.2.4 What's missing

We can get X (majoranas 1 and 2 switched), Z (majoranas 2 and 3 switched), and Y = i XZ operations, but missing something like T (pi / 8 rotation)

[7] Sec. IV B and C

Something something non-topological

supplementary operations

"The fact that braiding only enacts $\pi/2$ rotations is the reason why this type of nonAbelian anyon does not enable universal topological quantum computation, as we discuss further in section IV" - P.32 [7]

3.3 Methods using the Kitaev chain

3.3.1 T-junction

The T-junction is a proposed architecture for allowing braiding with a 1D Kitaev chain. Simply put, the idea stems from connecting a 1D chain to the middle of another, producing a T-shape. In this geometry, one can in theory braid the Majoranas by altering the parameters of the Hamiltonian to rotate the edge states around eachother without touching.

This figure should contain a numbering of the sites and bonds as the code uses them; starting at 0 on the central site and counting outwards, starting with the left arm, then the right, then the bottom.

Exchange of two Majoranas

Pali x y z gates

Double switches = rotations? (Drawing from meeting with Joakim)

Network of wires

Josephson currents

Ex. effect on the Pauli-X gate

Smallest T-junction

Proving that one actually only needs $N = 7$ (or $L = 5$) to have 4 MZMs (Eivind said this was possible).

3.3.2 Quantum dots

3.3.3 Comparison of T-junction and Quantum dots

3.3.4 Others

3.4 Topological protection

Chapter 4

Numerical methods

In this thesis we have numerically simulated the Kitaev chain T-junction. We have chosen to do so using the adiabatic approximation in hopes of reducing the otherwise overwhelming demand for computational power, aiming to truncate the state space while preserving numerical accuracy. In order to test the accuracy of the adiabatic approach we have also simulated the system by direct diagonalization for comparison. For integration we have used RK4 as it is relatively simple and produces accurate results, although it is not symplectic meaning it can drift from the true result if the time step is not sufficiently small.

This chapter covers the theory behind the methods used. It is meant mainly as an instruction on how to replicate the results, and contains very little to no interesting information about the physics of the Kitaev chain. Most readers will find this section at best dry, and we have theorized it could function as a sleep aid.

4.1 Hilbert space and many-body basis

Considering a Kitaev chain with L sites we can use an occupation number basis

$$\{|n\rangle\}, \quad |n\rangle = |n_1, n_2, ..., n_L\rangle, \quad n_i \in \{0, 1\}. \tag{4.1}$$

Where 0 represents a site being empty, and 1 represents a site being occupied. This defines our Hilbert space, with a corresponding Hamiltonian $\mathcal{H}$. This allows us to produce a Hamiltonian matrix

$$H_{nm} = \langle n|\mathcal{H}|m\rangle. \tag{4.2}$$

4.2 Diagonalizing the Hamiltonian

Diagonalizing by the use of a unitary matrix U with elements c_{nk}

$$H = UDU^{-1} \tag{4.3}$$

where D is diagonal containing the eigenvalues E_k. The energy eigenstates are related to the basis by

$$|\psi_k\rangle = \sum_n c_{nk}|n\rangle. \tag{4.4}$$

We can expand any state in the energy eigenbasis

$$|\phi\rangle = \sum_k a_k |\psi_k\rangle. \tag{4.5}$$

For a time-independent H, the time evolution of U can be written

$$U(t) = e^{-iHt} \tag{4.6}$$

such that

$$\begin{aligned} |\phi(t)\rangle &= U(t)|\phi\rangle && (4.7)\\ &= e^{-iHt}|\phi(0)\rangle && (4.8)\\ &= \sum_k a_k(0) e^{-iHt}|\psi_k\rangle && (4.9)\\ &= \sum_k a_k(t)|\psi_k\rangle && (4.10) \end{aligned}$$

Where $|\phi(0)\rangle = |\phi\rangle, a_k(0) = a_k$ from earlier. We see that the coefficients just pick up time-dependent phases

$$a_k(t) = a_k(0)e^{-iHt} \tag{4.11}$$

4.3 Time-dependent Hamiltonian & adiabatic basis

We now want to define a time dependent Hamiltonian with an orthonormal instantaneous basis

$$\begin{aligned} H(t); \quad \{|\psi_k(t)\rangle\} && (4.12)\\ H(t)|\psi_k(t)\rangle = E_k(t)|\psi_k(t)\rangle. && (4.13)\\ && (4.14) \end{aligned}$$

Expanding the full state as

$$|\phi(t)\rangle = \sum_k a_k(t) e^{-i\int_0^t dt' \lambda_k(t')}|\psi_k(t)\rangle \tag{4.15}$$

we can now determine the time evolution of the system using Schrödingers equation

$$\begin{aligned} -iH|\phi\rangle &= \frac{d}{dt}|\phi\rangle && (4.16)\\ -iH\sum_k a_k e^{-i\int_0^t \lambda_k}|\psi_k\rangle &= \frac{d}{dt}\sum_k a_k(t) e^{-i\int_0^t dt' \lambda_k(t')}|\psi_k(t)\rangle && (4.17)\\ -i\sum_k a_k e^{-i\int_0^t \lambda_k} E_k|\psi_k\rangle &= \sum_k \left[\frac{da_k}{dt}e^{-i\int_0^t \lambda_k}|\psi_k\rangle + a_k(-i\lambda_k)e^{-i\int_0^t \lambda_k}|\psi_k\rangle + a_k e^{-i\int_0^t \lambda_k}\frac{d|\psi_k\rangle}{dt}\right] && (4.18) \end{aligned}$$

$$\begin{aligned} -i\sum_k e^{-i\int_0^t \lambda_k}\left[a_k E_k|\psi_k\rangle - i\frac{da_k}{dt}|\psi_k\rangle - a_k\lambda_k|\psi_k\rangle - i a_k\frac{d|\psi_k\rangle}{dt}\right] = 0 && (4.19)\\ \sum_k e^{-i\int_0^t \lambda_k}\left[a_k\left((E_k-\lambda_k)|\psi_k\rangle - i\frac{d|\psi_k\rangle}{dt}\right) - i\frac{da_k}{dt}|\psi_k\rangle\right] = 0 && (4.20) \end{aligned}$$

We now project onto $\langle\psi_l|$ as we by orthonormality have $\langle\psi_l|\psi_k\rangle = \delta_{lk}$.

$$\langle\psi_l| \sum_k e^{-i\int_0^t \lambda_k} \left[a_k \left((E_k - \lambda_k)|\psi_k\rangle - i\frac{d|\psi_k\rangle}{dt} \right) - i\frac{da_k}{dt}|\psi_k\rangle \right] = 0 \tag{4.21}$$

$$\sum_k e^{-i\int_0^t \lambda_k} \left[a_k \left((E_k - \lambda_k)\delta_{l,k} - i\langle\psi_l|\frac{d}{dt}|\psi_k\rangle \right) - i\frac{da_k}{dt}\delta_{l,k} \right] = 0 \tag{4.22}$$

$$e^{-i\int_0^t \lambda_l} \left[a_l(E_l - \lambda_l) - i\frac{da_l}{dt} \right] - i\sum_k a_k e^{-i\int_0^t \lambda_k} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = 0 \tag{4.23}$$

$$a_l(E_l - \lambda_l) - i\frac{da_l}{dt} - ie^{i\int_0^t \lambda_l} \sum_k a_k e^{-i\int_0^t \lambda_k} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = 0 \tag{4.24}$$

$$\sum_k a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle - ia_l(E_l - \lambda_l) = \frac{da_l}{dt} \tag{4.25}$$

Remembering that λ is just a definition used to split off a phase from a_k, we can choose it to help simplify our equation. We see that by choosing

$$\lambda_k = E_k - i\langle\psi_k|\frac{d}{dt}|\psi_k\rangle \tag{4.26}$$

we can remove all diagonal terms.

$$\sum_k a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle - ia_l(E_l - E_l - i\langle\psi_l|\frac{d}{dt}|\psi_l\rangle) = \frac{da_l}{dt} \tag{4.27}$$

$$\sum_{k\neq l} a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle + a_l e^0 \langle\psi_l|\frac{d}{dt}|\psi_l\rangle - a_l\langle\psi_l|\frac{d}{dt}|\psi_l\rangle = \frac{da_l}{dt} \tag{4.28}$$

$$\sum_{k\neq l} a_k e^{-i\int_0^t (\lambda_l - \lambda_k)} \langle\psi_l|\frac{d}{dt}|\psi_k\rangle = \frac{da_l}{dt} \tag{4.29}$$

4.4 RK4 integration

Accuracy

4.5 Code structure

I used UiOGPT to help debug the code.

4.6 Setting up the Kitaev chain Hamiltonian matrix

Using original Hamiltonian 2.1.

Open boundary conditions

Hilbert space basis

Making H_{nm}

For mu term: Use $\sum_j n_j|n\rangle = \sum_j a_j^\dagger a_j|n\rangle = N|n\rangle$

For omega and delta terms: $c_j^\dagger c_{j+1}|n\rangle = c_1 c_2|m_2\rangle$

Use time dependent μ for time dependency.

4.7 Finding eigenvectors and eigenvalues by diagonalization

numpy.linalg.eigh

4.8 Calculating $\langle\psi_l|\frac{d}{dt}|\psi_k\rangle$ & λ

Finite difference (Numerical derivation) for the bulk $n \in [1, N-1]$

$$\langle\psi_l(t_n)|\frac{d}{dt}|\psi_k(t_n)\rangle = \frac{\langle\psi_l(t_n)|\psi_k(t_{n+1})\rangle - \langle\psi_l(t_n)|\psi_k(t_{n-1})\rangle}{2\Delta t}. \tag{4.30}$$

And a forward difference for $n = 0$

$$\langle\psi_l(t_0)|\frac{d}{dt}|\psi_k(t_0)\rangle = \frac{d}{dt}\left[\langle\psi_l(t_0)|\psi_k(t)\rangle\right]_{t_0} \tag{4.31}$$

$$= \frac{\langle\psi_l(t_0)|\psi_k(t_1)\rangle - \langle\psi_l(t_0)|\psi_k(t_0)\rangle}{t_1 - t_0} \tag{4.32}$$

$$= \frac{\langle\psi_l(t_0)|\psi_k(t_1)\rangle - \delta_{lk}}{dt}. \tag{4.33}$$

And a backward difference to find $n = N - 1$.

4.9 Time evolution of amplitudes

Using Euler.
Upgrade to RK4 later.

4.10 Evolving the fixed basis

Using H(t) with $\mu(t), \omega(t), \Delta(t)$
Here one can also evolve the ... basis as seen in Eivinds master

Chapter 5

Results & discussion

Eivind the GOAT

What do plots in the adiabatic basis actually tell us?

5.1 Splitting the chain and moving MZMs

5.1.1 Replicating results

A previous masters student also did this. Boom i replicated his results life is good

Numerical accuracy of the adiabatic approach

Truncation

Time and accuracy wrt truncation Do trunc = 4, 10, 150, 250?

1

1

1

1

1

Direct diagonalization vs. truncation

5.1.2 Moving MZMs along the chain

Checking that moving is not problematic - Use same times as used in T-junction already

5.2 T-junction

I need a supercomputer :/

5.2.1 Moving around intersection

Josepshon current issues

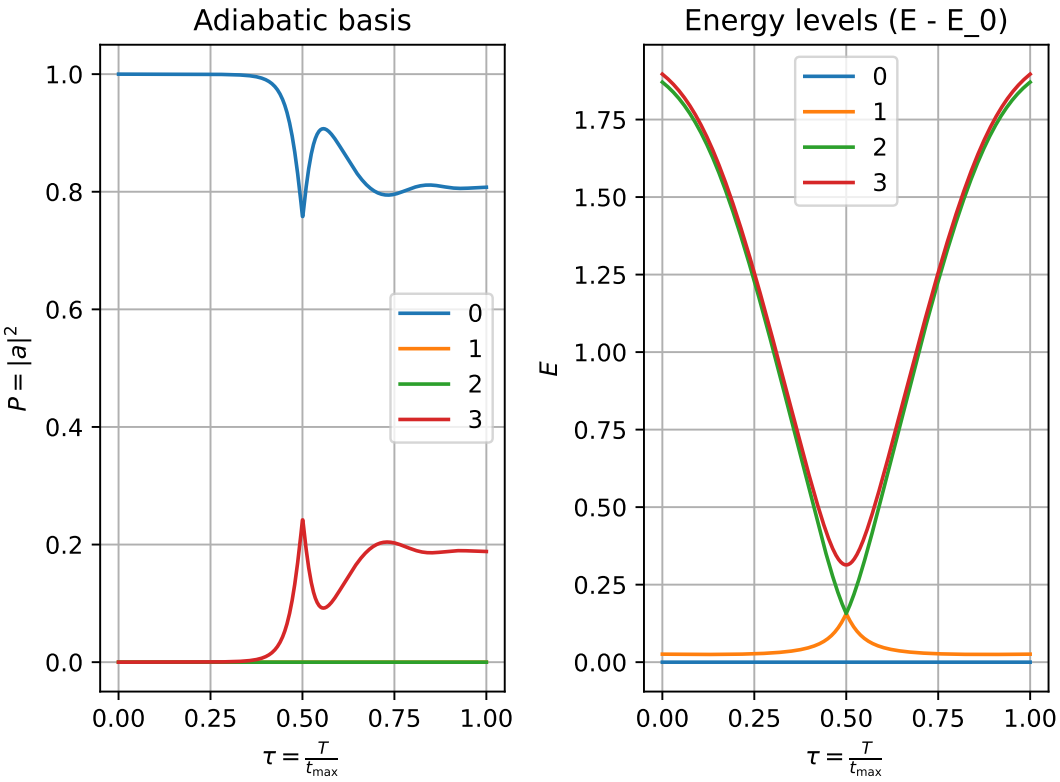


Figure 5.1: The 4 lowest energy states from the time evolution ($T = 20$) of a single chain of $N = 10$ sites truncated to the 150 lowest energy states.

5.2.2 Pauli-X gate

As previously discussed 3.3.1, the Pauli-X gate is incorporated by exchanging MZMs 2 and 3 as seen in Figure 3.2. From previous results 5.1.2 we know that moving the MZMs along a chain by a time-dependent chemical potential $\mu(\tau)$ is unproblematic as long as μ is changed slowly enough. And moving them through the intersection is unproblematic as long as the superconducting term Δ has the appropriate sign to avoid problems with the Josepshon current 5.2.1. Therefore, from a simulation standpoint, the only interesting part left of the Pauli-X gate operation is the central operation where MZMs 2 and 3 are switched through the central site. To simulate this we will be using $N = 10$ sites, and unless stated otherwise we will be using a truncation trunc $= 150$ and timestep $d\tau = 10^{-3}$. We follow the parameters used in [18]: Keeping the hopping term $\omega = 1$ and superconducting term $|\Delta| = 1$ constant, while moving the MZMs around by a time-dependent chemical potential

$$\mu_n(\tau) = \mu_{\text{triv}} + (\mu_{\text{topo}} - \mu_{\text{triv}})\, r(\tau), \tag{5.1}$$
$$\tag{5.2}$$

defined by a helper function [1].

$$r(\tau) = \begin{cases} 0 & \tau \leq 0 \\ \tau^2(3 - 2\tau) & 0 < \tau < 1 \\ 1 & \tau \geq 1 \end{cases}. \tag{5.3}$$

Giving

$$\rightarrow \mu_n(\tau) = \begin{cases} \mu_{\text{triv}} & \tau \leq 0 \\ \mu_{\text{triv}} + (\mu_{\text{topo}} - \mu_{\text{triv}})\, \tau^2(3 - 2\tau) & 0 < \tau < 1 \\ \mu_{\text{topo}} & \tau \geq 1. \end{cases} \tag{5.4}$$

Where $\mu_{\text{triv}} = 8, \mu_{\text{topo}} = 0.05$. These values allow us to perform our desired simulations, and also allow us to cross-check with their findings if relevant.

The parameters of this simulation are visualized in Figure 5.2. The site and bond index follow Figure 3.1. This shows a constant hopping parameter and superconducting term, with the superconducting term being negative on the vertical arm. The interesting part of the figure is $\mu(j, \tau)$, showing how the central site is always in the non-trivial phase. While the first site on the left arm slowly goes from the non-trivial to the trivial phase, and the first site on the right arm the does opposite.

Our theoretical understanding of the appropriate sign for Δ to avoid problems with the Josepshon current was only earned after initial testing with the code. Initially we used $\Delta = 1$ for the whole T-junction, and we will therefore also be discussing these results.

Equal sign on the superconducting term for the whole T-junction

When using $\Delta = 1$ for the whole chain we get 8 degeneracies of the ground state, which is unexpected. Although these 8 states remain degenerate throughout the time evolution this is unexpected as it means there are 8 MZMs for two topological chains. We believe

[1] This is not exactly the definition [18] used; their calling of $r(\tau)$ also allows for a spatially staggered site-by-site ramp using a parameter α. This is however only of use when $N > 10$ as for $N = 10$ we only ramp 1 site on each arm.

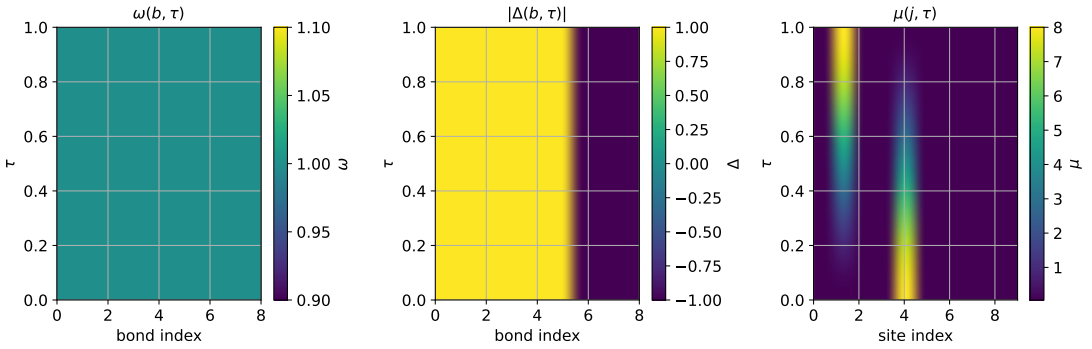


Figure 5.2: Time dependent parameters for time evolution of the switching of Majoranas 2 and 3 as shown in Figure 3.2. The whole T-junction has $\omega = 1$. The horizontal arms have $\Delta = 1$, whereas the vertical arm has $\Delta = -1$. μ is defined by equation 5.4, using $\mu_{\text{triv}} = 8$, $\mu_{\text{topo}} = 0.05$.

this is due to Josephson currents effectively cutting the T-junction into separate chains at the central site.

In light of the unexpected results of the degenerations mentioned above we check the energy spectrum of the chain when the whole T-chain is in the topological phase $\mu = \mu_{\text{topo}}$. When attempting $\Delta = 1$ for the whole chain we get an unnormalized ground state probability $|a_0|^2 \approx 60$, and $E_0 = 0, E_{\text{all other}} = -1$. This is very much unphysical, and is evidence something is horribly wrong. This result was made with a high truncation trunc $= 500$, as this is only the first timestep of the simulation, and is as such much less computationally expensive to produce. Further, a high truncation provides a lot more detailed picture of the energy spectrum [2].

Appropriate superconducting term

When however using an appropriate superconducting term $\Delta = 1$ for the horizontal arms, and $\Delta = -1$ for the vertical arm, again with $\mu = \mu_{\text{topo}}$, we get the energy spectrum as shown in Figure 5.3. Here we have a four-fold degeneracy of the lowest energy state with energies $E = 0, 8.496 \cdot 10^{-12}, 3.123 \cdot 10^{-5}, 3.123 \cdot 10^{-5}$. This is as expected, and points to the simulation producing physical results for the parameters given.

Running the full time-dependent simulation with the appropriate superconducting term we get Figure 5.4. We here see that we get the expected four-fold degeneration of the ground state, and there is no energy splitting of the ground state during the time evolution. The energy also forms energy bands which are symmetrical about $\tau = 0.5$, which is expected as the Hamiltonian returns to its original form as $\tau \to 1$. For higher energy states we observe discontuity. We believe this to be a numerical error which is mostly only problematic during visualization. The energy bands typically consist of degenerate energy levels which are difficult to keep track of numerically as the system evolves and we find new energy levels at each time step. Further we can see that the system starts in the ground state in the adiabatic basis, before ending with nearly all probability split between the first and the third state, with a near 0% probability of finding the system in any other state.

If we observe an energy splitting we expect the stored information to vanish and as such we wish to test how the Majorana zero modes respond to a different time

[2] In hindsight we probably should've kept trunc $= 150$ for consistency, and trunc $= 500$ being unnecessarily high. Using trunc $= 150$ is however highly unlikely to change the conclusion and as such we've kept these results.

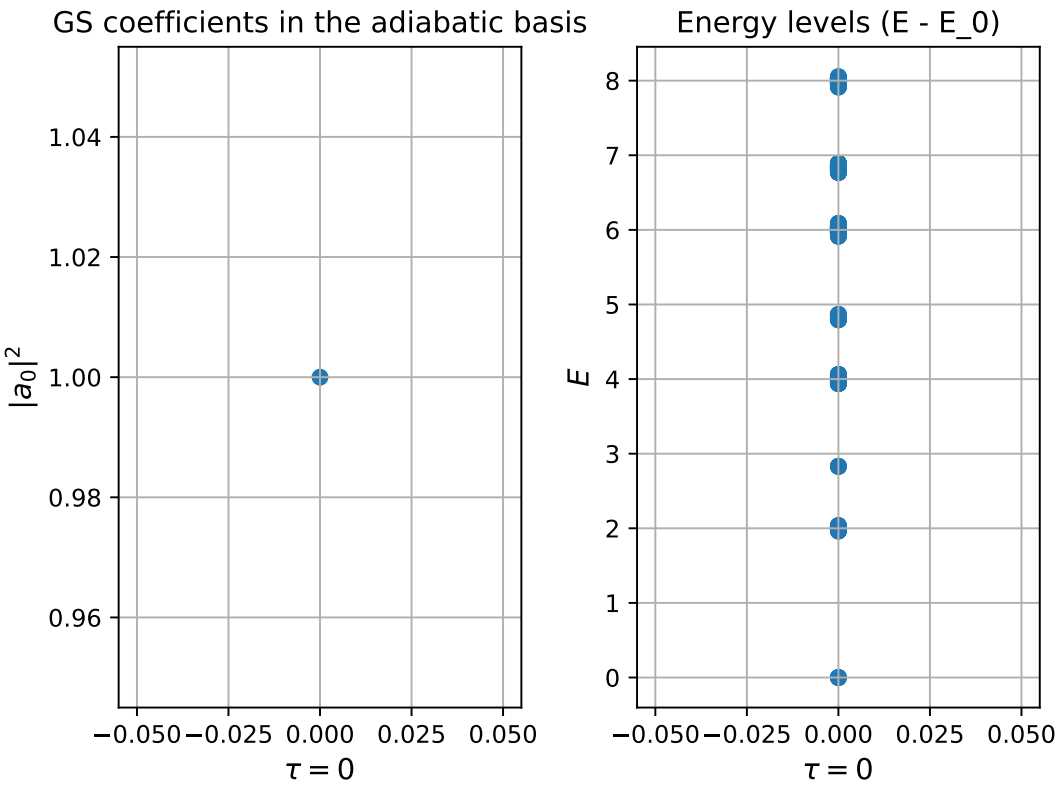


Figure 5.3: Initial ground state coefficient and energy spectrum for the fully topological T-chain ($\omega = 1, \mu = 0.05$) for the appropriate superconducting term where the vertical arm has $\Delta = -1$, and the horizontal arms have $\Delta = 1$.

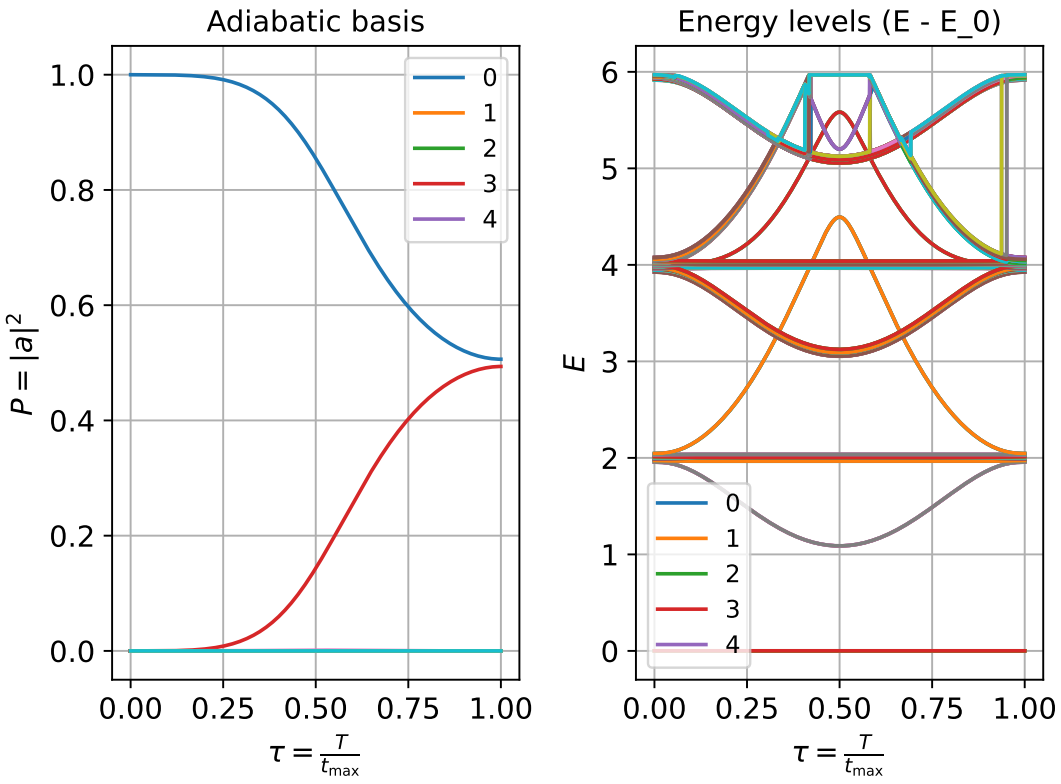


Figure 5.4: Probability of the adiabatic basis and the energy levels (subtracted the ground state) for time evolution of the switching of Majoranas 2 and 3 in the central operation of the Pauli-X gate. Using $N = 10$, showing all trunc $= 150$ states, using $T = 20$, and the appropriate parameters $\omega = 1|\Delta| = 1$ with $\mu(j, \tau)$ as in equation 5.3. The parameters are visualized in Figure 5.2. The ground state energy is four-fold degenerate.

evolution of the operation. As such we use $\omega = |\Delta| = 1$ (again as above), with $\{\mu_{\text{triv}}, \mu_{\text{topo}}\} = \{\{0.05, 5\}, \{0.1, 8\}\}$... ... ...

5.2.3 Pauli Z-gate

Exchange 1 and 2 (or 3 and 4)

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Appendix A

Calculations

A.1 Transforming the Kitaev chain Hamiltonian to Majorana operators

This appendix shows how transform the original Kitaev chain Hamiltonian expressed in fermionic operators a_j and $a_j^\dagger$ to being expressed in terms of the majorana operators c_{2j-1} and $c_2 j$.

Starting with the majorana operators

$$c_{2j-1} = e^{i\frac{\theta}{2}} a_j + e^{-i\frac{\theta}{2}} a_j^\dagger \tag{A.1}$$
$$c_{2j} = -ie^{i\frac{\theta}{2}} a_j + ie^{-i\frac{\theta}{2}} a_j^\dagger. \tag{A.2}$$

We can invert them, leading to four expressions

$$a_j = e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger \tag{A.3}$$
$$= ie^{-i\frac{\theta}{2}} c_{2j} + e^{-i\theta} a_j^\dagger \tag{A.4}$$
$$a_j^\dagger = e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j \tag{A.5}$$
$$= -ie^{i\frac{\theta}{2}} c_{2j} + e^{i\theta} a_j. \tag{A.6}$$

Combining the expressions for a_j, and for $a_j^\dagger$ we find

$$a_j = \frac{1}{2}\left(e^{-i\frac{\theta}{2}} c_{2j-1} - e^{-i\theta} a_j^\dagger + e^{-i\theta} a_j^\dagger + ie^{-i\frac{\theta}{2}} c_2 j\right) \tag{A.7}$$
$$= \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + ic_{2j}\right) \tag{A.8}$$
$$a_j^\dagger = \frac{1}{2}\left(e^{i\frac{\theta}{2}} c_{2j-1} - e^{i\theta} a_j + e^{i\theta} a_j - ie^{i\frac{\theta}{2}} c_{2j}\right) \tag{A.9}$$
$$= \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - ic_{2j}\right). \tag{A.10}$$

We now substitute these expressions into the original Hamiltonian

$$
\begin{aligned}
H =& \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] && \text{(A.11)} \\
=& -\omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) && \text{(A.12)} \\
& - \omega \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) && \text{(A.13)} \\
& - \mu \sum_j \left(\frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) - \frac{1}{2} \right) && \text{(A.14)} \\
& + \Delta \sum_j \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2j-1} + i c_{2j} \right) \frac{1}{2} e^{-i\frac{\theta}{2}} \left(c_{2(j+1)-1} + i c_{2(j+1)} \right) && \text{(A.15)} \\
& + \Delta^* \sum_j \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2(j+1)-1} - i c_{2(j+1)} \right) \frac{1}{2} e^{i\frac{\theta}{2}} \left(c_{2j-1} - i c_{2j} \right) && \text{(A.16)} \\
=& -\frac{\omega}{4} \sum_j \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} - i c_{2j} c_{2j+1} + c_{2j} c_{2j+2} \right) && \text{(A.17)} \\
& - \frac{\omega}{4} \sum_j \left(c_{2j+1} c_{2j-1} + i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} + c_{2j+2} c_{2j} \right) && \text{(A.18)} \\
& - \frac{\mu}{4} \sum_j \left(c_{2j-1}^2 + i c_{2j-1} c_{2j} - i c_{2j} c_{2j-1} + c_{2j}^2 - 2 \right) && \text{(A.19)} \\
& + \frac{|\Delta|}{4} e^{i\theta} \sum_j e^{-i\theta} \left(c_{2j-1} c_{2j+1} + i c_{2j-1} c_{2j+2} + i c_{2j} c_{2j+1} - c_{2j} c_{2j+2} \right) && \text{(A.20)} \\
& + \frac{|\Delta|}{4} e^{-i\theta} \sum_j e^{i\theta} \left(c_{2j+1} c_{2j-1} - i c_{2j+1} c_{2j} - i c_{2j+2} c_{2j-1} - c_{2j+2} c_{2j} \right) && \text{(A.21)} \\
=& -\frac{\omega}{4} \sum_j \left(0 + 2i c_{2j-1} c_{2j+2} - 2i c_{2j} c_{2j+1} + 0 \right) && \text{(A.22)} \\
& - \frac{\mu}{4} \sum_j \left(2i c_{2j-1} c_{2j} + 2 - 2 \right) && \text{(A.23)} \\
& + \frac{|\Delta|}{4} \sum_j \left(0 + 2i c_{2j-1} c_{2j+2} + 2i c_{2j} c_{2j+1} + 0 \right) && \text{(A.24)} \\
=& \frac{i}{2} \sum_j \left[-\mu c_{2j-1} c_{2j} + \left(\omega + |\Delta| \right) c_{2j} c_{2j+1} + \left(-\omega + |\Delta| \right) c_{2j-1} c_{2j+2} \right] && \text{(A.25)}
\end{aligned}
$$

Where we used $\Delta = |\Delta| e^{i\theta}$ and the anticommutation relations

$$
\begin{aligned}
\{ c_l, c_m \} = c_l c_m + c_m c_l &= 2\delta_{lm} && \text{(A.26)} \\
c_l c_m &= -c_m c_l (m \neq l) && \text{(A.27)} \\
c_l^2 &= 1. && \text{(A.28)}
\end{aligned}
$$

A.2 Transforming the Kitaev chain Hamiltonian to momentum space

Taking inspiration from [15] we start from the Kitaev chain Hamiltonian

$$H = \sum_j \left[-\omega \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) - \mu \left(a_j^\dagger a_j - \frac{1}{2} \right) + \Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right] \tag{A.29}$$

Imposing periodic boundary conditions we can Fourier transform the Hamiltonian to momentum space q. We recall the Fourier transform

$$a_j = \frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q, \quad a_j^\dagger = \frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \tag{A.30}$$

where $q = \frac{2\pi n}{N}$, with n such that $q$runs over the first Brillouin zone $[-\pi, \pi]$. These operators keep the anticommutation relations of the real-space operators

$$\{a_q, a_{q'}\} = \frac{1}{N} \sum_{jj'} e^{-iqj} e^{-iq'j'} \{a_j, a_{j'}\} = 0 \tag{A.31}$$

$$\left\{a_q, a_{q'}^\dagger\right\} = \frac{1}{N} \sum_{jj'} e^{-iqj} e^{iq'j'} \left\{a_j, a_{j'}^\dagger\right\} \tag{A.32}$$

$$= \frac{1}{N} \sum_{jj'} e^{-iqj} e^{iq'j'} \delta_{j,j'} \tag{A.33}$$

$$= \frac{1}{N} \sum_{j} e^{i(q'-q)j} \tag{A.34}$$

$$= \delta_{q,q'} \tag{A.35}$$

Using this we can transform each term in the Hamiltonian individually for simplicity. Starting with the on-site term

$$H_\mu = -\mu \sum_j \left(a_j^\dagger a_j - \frac{1}{2} \right) \tag{A.36}$$

$$= -\mu \sum_j \left(\left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'j} a_{q'} \right) - \frac{1}{2} \right) \tag{A.37}$$

$$= -\mu \frac{1}{N} \sum_{jqq'} e^{i(q'-q)j} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.38}$$

$$= -\frac{\mu}{N} \sum_{qq'} N \delta_{q,q'} a_q^\dagger a_{q'} + \frac{\mu N}{2} \tag{A.39}$$

$$= -\mu \sum_q a_q^\dagger a_q + \frac{\mu N}{2}. \tag{A.40}$$

Where we used a property of the periodic boundary conditions $\sum_j e^{i(q'-q)j} = N\delta_{q,q'}$. Continuing with the hopping term

$$H_\omega = -\omega \sum_j \left(a_j^\dagger a_{j+1} + a_{j+1}^\dagger a_j \right) \tag{A.41}$$

where we again split the calculation of the terms; the first term becomes

$$H_\omega^1 = -\omega \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{-iqj} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'} \right) \tag{A.42}$$

$$= -\frac{\omega}{N} \sum_{jqq'} e^{i(q'-q)j} e^{iq'} a_q^\dagger a_{q'} \tag{A.43}$$

$$= -\frac{\omega}{N} \sum_{qq'} N \delta_{q,q'} e^{iq'} a_q^\dagger a_{q'} \tag{A.44}$$

$$= -\omega \sum_q e^{iq} a_q^\dagger a_q. \tag{A.45}$$

And the second term similarly becomes

$$H_\omega^2 = -\omega \sum_q e^{-iq} a_q^\dagger a_q. \tag{A.46}$$

Combining the hopping terms we get

$$H_\omega = H_\omega^1 + H_\omega^2 \tag{A.47}$$

$$= -\omega \sum_q \left(e^{iq} + e^{-iq} \right) a_q^\dagger a_q \tag{A.48}$$

$$= -2\omega \sum_q \cos(q) a_q^\dagger a_q. \tag{A.49}$$

We now consider the pairing term

$$H_\Delta = \sum_j \left(\Delta a_j a_{j+1} + \Delta^* a_{j+1}^\dagger a_j^\dagger \right) \tag{A.50}$$

$$\tag{A.51}$$

where we again split the calculation for simplicity

$$H_\Delta^1 = \Delta \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{iqj} a_q \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{iq'(j+1)} a_{q'} \right) \tag{A.52}$$

$$= \frac{\Delta}{N} \sum_{jqq'} e^{i(q+q')j} e^{iq'} a_q a_{q'} \tag{A.53}$$

$$= \frac{\Delta}{N} \sum_{qq'} N \delta_{q,-q'} e^{iq'} a_q a_{q'} \tag{A.54}$$

$$= \Delta \sum_q e^{-iq} a_q a_{-q}. \tag{A.55}$$

And the second term similarly becomes

$$H_\Delta^2 = \Delta^* \sum_j \left(\frac{1}{\sqrt{N}} \sum_q e^{-iq(j+1)} a_q^\dagger \right) \left(\frac{1}{\sqrt{N}} \sum_{q'} e^{-iq'j} a_{q'}^\dagger \right) \tag{A.56}$$

$$= \frac{\Delta^*}{N} \sum_{jqq'} e^{-i(q+q')j} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.57}$$

$$= \frac{\Delta^*}{N} \sum_{qq'} N \delta_{q,-q'} e^{-iq} a_q^\dagger a_{q'}^\dagger \tag{A.58}$$

$$= \Delta^* \sum_q e^{-iq} a_q^\dagger a_{-q}^\dagger. \tag{A.59}$$

Combining all terms we get the momentum-space Hamiltonian

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q + \Delta e^{-iq} a_q a_{-q} + \Delta^* e^{-iq} a_q^\dagger a_{-q}^\dagger \right] + \frac{\mu N}{2} \tag{A.60}$$

Where the last term $+\frac{\mu N}{2}$ can be ignored as it does not affect the spectrum, eigenstates, or any observable that depends on the energy differences. As such it will be disregarded for the rest of the thesis. Later we will use this expression to construct the BdG Hamiltonian, which requires the pairing terms to be in an antisymmetric form, which they are not currently in. We can create the antisymmetric form by expanding the operators to be a combination of both orderings $a_q a_{-q}, a_{-q} a_q$ (and similar for the † terms) by the anticommutation relations. We start by considering the annihilation term

$$\sum_q e^{-iq} a_q a_{-q} = \frac{1}{2} \sum_q \left[e^{-iq} a_q a_{-q} + e^{-iq} a_q a_{-q} \right] \tag{A.61}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_q e^{-iq} a_{-q} a_q \right] \tag{A.62}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_{-q} e^{iq} a_q a_{-q} \right] \tag{A.63}$$

$$= \frac{1}{2} \left[\sum_q e^{-iq} a_q a_{-q} - \sum_q e^{iq} a_q a_{-q} \right] \tag{A.64}$$

$$= \frac{1}{2} \sum_q \left[e^{-iq} - e^{iq} \right] a_q a_{-q} \tag{A.65}$$

$$= \frac{1}{2} \sum_q \left[-2i \sin(q) \right] a_q a_{-q} \tag{A.66}$$

$$= -i \sum_q \sin(q) a_q a_{-q} \tag{A.67}$$

$$\tag{A.68}$$

Where we also renambed the dummy variable $q \to -q$ for the second term. We are allowed to do this as a sum over q is equivalent to a sum over $-q$ by the periodic boundary conditions. The corresponding calculation for the creation term similarly gives

$$\sum_q e^{-iq} a_q^\dagger a_{-q}^\dagger = -i \sum_q \sin(q) a_q^\dagger a_{-q}^\dagger. \tag{A.69}$$

As such the final momentum-space Hamiltonian with antisymmetric pairing terms is

$$H = \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q - i\Delta \sin(q) a_q a_{-q} - i\Delta^* \sin(q) a_q^\dagger a_{-q}^\dagger \right] \tag{A.70}$$

$$= \sum_q \left[-\mu a_q^\dagger a_q - 2\omega \cos(q) a_q^\dagger a_q - i \sin(q) \left(\Delta a_q a_{-q} + \Delta^* a_q^\dagger a_{-q}^\dagger \right) \right] \tag{A.71}$$

A.3 Finding the evolution of the system under the adiabatic assumption

Move huge ugly calculation from methods to here(?)